

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

D. Madhusudhan
192524214
D. M.

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies 25	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues 25	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration 20	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results 15	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results 10	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

95
100



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

6G Communication Architecture Compared with the OSI Model

CSA0720-COMPUTER NETWORKS

Guided by

Dr.AGNES SHIFFANI

Presented by

D.MADHUSUDHAN(192524214)

T.TRIVIKRAM(192524217)



INTRODUCTION



- **What is 6G?**
- 6G stands for **Sixth Generation Wireless Communication**.
- Expected to be introduced around **2030**.
- Provides **ultra-fast speed** and **very low latency**.
- Uses **Artificial Intelligence (AI)** for smarter communication.
- Supports advanced technologies like:
 - Internet of Things (IoT)
 - Smart Cities
 - Autonomous Vehicles
 - Holographic Communication
 - Extended Reality (XR)



Key Features

- Speed up to **1 Tbps**
- Latency less than **1 millisecond**
- AI-powered network management
- High reliability
- Better security
- Energy-efficient communication
- Global connectivity including satellites



OSI Model



- The OSI Model has **7 layers**:
- Physical Layer
- Data Link Layer
- Network Layer
- Transport Layer
- Session Layer
- Presentation Layer
- Application Layer



6G Communication Architecture



- **Main Components**

- User Devices
- Base Stations
- Edge Computing
- AI Controller
- Core Network
- Cloud Services
- Satellite Networks

- **Working**

- Device → Base Station → Edge Computing → Core Network → Internet/Cloud



Comparison of 6G with OSI Layers



OSI Layer

Physical

Data Link

Network

Transport

Session

Presentation

Application

Role in 6G

THz wireless communication

Error detection and reliable transmission

AI-based intelligent routing

Fast and reliable data delivery

Maintains communication sessions

Encryption and data formatting

Smart applications like IoT and AR/VR



Applications of 6G



- **Applications**
- Smart Cities
- Smart Healthcare
- Self-driving Cars
- Industrial Automation
- Smart Agriculture
- Virtual Reality (VR)
- Augmented Reality (AR)
- Holographic Communication

Advantages and Challenges

- **Advantages**

- Very high speed
- Low latency
- Better security
- AI-based communication
- Energy efficient

- **Challenges**

- High deployment cost
- Infrastructure requirements
- Security risks
- Privacy concerns
- Spectrum availability

Conclusion

- 6G is the future of wireless communication.
- It provides faster, smarter, and more secure communication.
- The OSI Model remains the foundation of networking.
- Every OSI layer is enhanced using AI and advanced technologies in 6G

Future Scope

- **Future of 6G**
- AI-driven intelligent networks
- Global satellite communication
- Digital Twin technology

*Thank
you!*

 **SIMATS**
ENGINEERING

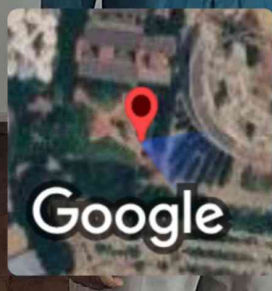
 **SIMATS**
Sreevidya Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

6G Communication Architecture Compared with the OSI Model

CSA0720-COMPUTER NETWORKS

Guided by
Dr.AGNES SHIFFANI

Presented by
D.MADHUSUDHAN(192524214)
T.TRIVIKRAM(192524217)



Mevalurkuppam, Tamil Nadu, India



22g8+c47, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.025915° Long 80.01553°
Thursday, 06/08/2026 09:16 AM GMT +05:30

 GPS Map Camera